

Stethoscope Sprint and Recovery

Build a medical listening tool and use it like a sports scientist.

Win condition: Best sound quality plus accurate, respectful data collection.

THE SCIENCE YOU ARE USING

Sound transmission. A stethoscope collects faint body sounds with a wide funnel and channels them through a tube to your ear. A sealed air path and a good funnel make quiet heartbeats audible. Materials and seal quality decide how well it works.

Heart rate and recovery. After light exercise, heart rate rises and then recovers. With consent, you measure resting and post-activity rates and watch recovery. Respectful, accurate data collection is part of the score.

HOW TO BUILD AND RUN IT

- 01 **Build the stethoscope.** Attach a funnel to tubing and seal the joints so no air leaks.
- 02 **Add a diaphragm.** Stretch a balloon membrane over the funnel to pick up faint sounds, optional.
- 03 **Test sound quality.** Listen for a clear heartbeat. Improve the seal until it is easy to hear.
- 04 **Collect resting data.** With consent, count the radial pulse at the wrist for a timed window. If the stethoscope is hard to hear, the wrist count still works. Do not take a neck pulse.
- 05 **Measure recovery.** After optional light activity, measure how heart rate trends toward its starting level. Anyone may opt out and take a recorder or timer role. Stop immediately and tell an adult if you feel dizzy, have chest pain, have unusual trouble breathing, or feel unwell. Sanitize shared parts.

Safety: Sanitize shared parts. Wrist pulse only; no neck pulse. Measurement and light activity are optional. Follow any health-plan or staff restriction, and stop immediately for dizziness, chest pain, unusual trouble breathing, or feeling unwell. Allergy: Latex balloon alternative: plastic wrap or nitrile membrane.

MATERIALS

Funnels	per team
Flexible tubing	per team
Tape	shared
Balloons or membranes	shared
Stopwatches	per team
Sanitizing wipes	shared

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): _____

The ONE variable we are testing: _____

Baseline result: _____

After redesign: _____

DEFEND YOUR DESIGN

What evidence made you change your design? _____

If you ran it again, what is the ONE thing you would change? _____

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Sound quality	30
Measurement accuracy	25
Explanation	20
Redesign	15
Data clarity	10
Total	100